

Yijie Xu 徐亦捷

Ph.D. Candidate in Artificial Intelligence at HKUST(GZ)

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RESEARCH INTEREST

His current research centers on how language models acquire and leverage information to adapt and make decisions, with a focus on **test-time self-evolution**, **generative retrieval**, and **retrieval-augmented generation (RAG)**.

EDUCATION

Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

PH.D. CANDIDATE IN ARTIFICIAL INTELLIGENCE

Aug. 2027 Expected

- **Supervisor:** Prof. Hui Xiong, Associate Vice President, Founding Head of AI Thrust, Fellow of the ACM, IEEE, AAAS, AAAI, and CAAI.
- **Co-supervisor:** Prof. Xuming Hu, Assistant Professor in AI Thrust, HKUST(GZ).

Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

M.PHIL. IN ARTIFICIAL INTELLIGENCE

Sep. 2022 - Aug. 2024

- **CGPA:** 3.96/4.3
- **Supervisor:** Prof. Hui Xiong, Associate Vice President, Founding Head of AI Thrust, Fellow of the ACM, IEEE, AAAS, AAAI, and CAAI.
- **Co-supervisor:** Prof. Xiaojuan Ma, Associate Professor in CSE, HKUST.

University of Reading &

Reading, England / Nanjing, China

Nanjing University of Information Science and Technology

DUAL B.SC. IN MATHEMATICS AND APPLIED MATHEMATICS

Sep. 2018 - Jun. 2022

- **Dual-degree** program.
- **CGPA:** 3.71/4
- **Final Year Project:** Research on Interpolation and Data Fitting: Basis and Applications.

PUBLICATIONS

†: Co-first Authors; ‡: Co-second Authors; *: Corresponding Authors.

- [1] **Unsupervised Post-Training of Foundation Models: A Survey**
Yijie Xu, Qianyi Cai, Huizai Yao, Yili Wang, Tianfu Wang, Cehao Yang, Xingbo Yao, Zhiyu Guo, Aiwei Liu, Xuming Hu*, Weiyu Guo*, Hui Xiong*
EMNLP'26: *The 2026 Conference on Empirical Methods in Natural Language Processing*
- [2] **Against Political Polarization: A Unified Framework for Tracing Evolving Political Ideologies on Social Media**
Yijie Xu, Chao Wang*, Hui Xiong*
TIST: *ACM Transactions on Intelligent Systems and Technology*
- [3] **You only need 4 extra tokens: Synergistic Test-time Adaptation for LLMs**
Yijie Xu, Huizai Yao, Zhiyu Guo, Weiyu Guo, Pengteng Li, Aiwei Liu, Xuming Hu, Hui Xiong*
Under Review for NeurIPS'26
Honored to be cited by **Prof. Yann LeCun** and **Jitendra Malik** as a representative exploration of LLM test-time adaptation.
- [4] **Rethinking Test-Time Adaptation for LLMs as Decoupled Unsupervised Post-Training**
Yijie Xu, Yili Wang, Qianyi Cai, Huizai Yao, Hanqing Wang, Tianfu Wang, Aiwei Liu, Weiyu Guo, Xuming Hu*, Hui Xiong*
Under Review for NeurIPS'26
- [5] **SePer: Measure Retrieval Utility Through The Lens Of Semantic Perplexity Reduction**
Lu Dai, Yijie Xu, Jinhui Ye, Hao Liu*, Hui Xiong*
ICLR'25 **Spotlight(3.20%)**: *The Thirteenth International Conference on Learning Representations*
- [6] **Mark Your LLM: Detecting the Misuse of Open-Source Large Language Models via Watermarking**
Yijie Xu†, Aiwei Liu†, Xuming Hu*, Lijie Wen, Hui Xiong*
WMARK@ICLR2025: *The 1st Workshop on GenAI Watermarking, ICLR 2025*
- [7] **VL-Eraser: Vacuum Distillation for Machine Unlearning in Vision-Language Models**
Yili Wang, Lu Dai†, Tairan Huang†, Yijie Xu†, Hui Xiong*
CVPR'26: *The IEEE / CVF Computer Vision and Pattern Recognition Conference*
- [8] **SafeBuild-Bench: A Temporal-Robust Construction Safety Benchmark with Graph-Enhanced Data Mining**
Yi Cui†, Zilin Wang†, Yijie Xu, Qianyi Cai, Huizai Yao, Shuai Jiang, Bingzhuo Zhong*, Hui Xiong*
KDD Datasets & Benchmarks Track'26
- [9] **Revisiting Noise Resilience Strategies in Gesture Recognition: Short-Term Enhancement in sEMG Analysis**
Weiyu Guo, Ziyue Qiao, Ying Sun*, Yijie Xu, Hui Xiong*
ICML'25: *The Forty-Second International Conference on Machine Learning*

- [10] **Logic-in-Frames: Dynamic Keyframe Search via Visual Semantic-Logical Verification for Long Video Understanding**
Weiyu Guo, Ziyang Chen, Shaoguang Wang, Jianxiang He, Yijie Xu, Jinhui Ye, Ying Sun*, Hui Xiong*
NeurIPS'25: The Thirty-ninth Annual Conference on Neural Information Processing Systems
- [11] **See&Trek: Training-Free Spatial Prompting for Multimodal Large Language Model**
Pengteng Li, Pinhao Song, Wuyang Li, Huizai Yao, Weiyu Guo, Yijie Xu, Dugang Liu, Hui Xiong*
NeurIPS'25: The Thirty-ninth Annual Conference on Neural Information Processing Systems
- [12] **Beyond Boundaries: Leveraging Vision Foundation Models for Source-Free Object Detection**
Huizai Yao, Sicheng Zhao, Pengteng Li, Yi Cui, Shuo Lu, Weiyu Guo, Yunfan Lu, Yijie Xu, Hui Xiong*
AAAI'26: The Fortieth AAAI Conference on Artificial Intelligence
- [13] **Less is More: Token-Efficient Video-QA via Adaptive Frame-Pruning and Semantic Graph Integration**
Shaoguang Wang, Jianxiang He, Weiyu Guo, Ziyang Chen, Yijie Xu, Xuming Hu, Hui Xiong*
Findings of CVPR'26: The IEEE / CVF Computer Vision and Pattern Recognition Conference
- [14] **SpGesture: Source-Free Domain-adaptive sEMG-based Gesture Recognition with Jaccard Attentive Spiking Neural Network**
Weiyu Guo, Ying Sun*, Yijie Xu, Ziyue Qiao, Yongkui Yang, Hui Xiong*
NeurIPS'24: The Thirty-eighth Annual Conference on Neural Information Processing Systems
- [15] **Event Camera Demosaicing via Swin Transformer and Pixel-focus Loss**
Yunfan Lu, Yijie Xu, Wenzong Ma, Weiyu Guo, Hui Xiong*
CVPRW'24: Workshop for IEEE / CVF Conference on Computer Vision and Pattern Recognition
- [16] **MIPI 2024 Challenge on Demosaic for HybridEVS Camera: Methods and Results**
Yaqi Wu, *et al.*, Yunfan Lu, Yijie Xu, Wenzong Ma, Weiyu Guo, Hui Xiong, *et al.*
CVPRW'24: Workshop for IEEE / CVF Conference on Computer Vision and Pattern Recognition
- [17] **Gated Test-Time Scaling for GUI Grounding via Multi-Seed Zoom Search**
Huizai Yao, Yijie Xu, Tianfu Wang, Yili Wang, Yi Cui, Yunfan Lu, Fengjie Zhu, Xuemin Zhao, Hengyao Bao, Ying Sun*, Hui Xiong*
Under Review for ACM MM'26
- [18] **Machine Unlearning in Diffusion LLMs**
Yili Wang, Yijie Xu, Lu Dai, Tairan Huang, Qianyi Cai, Huizai Yao, Tianfu Wang, Hui Xiong*
Under Review for NeurIPS'26
- [19] **AutoSoP: Automated Software Profiling for GUI Agents**
Tianfu Wang, Linxiao Gong, Huizai Yao, Yunfan Lu, Yijie Xu, Yili Wang, Qianyi Cai, Fengjie Zhu, Xuemin Zhao, Hengyao Bao, Ying Sun, Hui Xiong*
Under Review for NeurIPS'26
- [20] **Two Drifts, One Principle: Conflict-Aware Spectral Consolidation for Multimodal Continual Learning**
Haiyi Zhang, Qianyi Cai, Hanqing Wang, Zilin Wang, Xinyue Di, Yijie Xu, Yili Wang, Tianfu Wang, Yifan Han, Hui Xiong*
Under Review for NeurIPS'26
- [21] **A Survey of Agent Skills: Toward Procedural Infrastructure for LLM Agents**
Cehao Yang, Xiaojun Wu, Honghao Liu, Xueyuan Lin, Chengjin Xu, Xuhui Jiang, Yuanliang Sun, Wenjie Zhang, Zhichao Shi, Yijie Xu, Jia Li, Hui Xiong*, Jian Guo*
Preprints
- [22] **Contrastive Graph Matching for Domain Adaptive Object Detection**
Pengteng Li, Huizai Yao, Yi Cui, Yunfan Lu, Yijie Xu, Minghao Yang, Pinhao Song
arXiv

SELECTED PROJECTS

- Construction Supervision: Regulation-Grounded LLM/VLM System** Sep. 2025 - Present / Poly Group
- **Screened** and structured evidence from ~17k legal and regulatory documents for construction supervision.
 - **Built** a full-stack RAG/VLM system for regulation-grounded assessment and visual construction-site hazard identification.
- Multimodal HCI Perception Modeling** Jan. 2023 - Aug. 2024 / MetaHCI
- **Led** platform setup, data collection, algorithm design, and group coordination for an RBM multimodal HCI project.
 - **Developed** LLM-for-AIGC workflows, including training, visualization, deployment, and service maintenance for MetaHCI, a HKUST(GZ)-affiliated startup valued at 70 million RMB.
- Political Ideology Analysis on Social Media** Nov. 2022 - Oct. 2023
- **Built** a 230k+ report dataset from millions of tweets and author profiles.
 - **Proposed** a NLP + Temporal Graph Neural Network framework for ideology detection and prediction.
- Big Data Mining Services for CRM Systems** Nov. 2022 - May. 2023 / Tencent
- **Built** web-crawling and pre-trained LM pipelines for large-scale academic talent profiling.
 - **Operationalized** extracted profiles for an industry-facing CRM project at Tencent.

SELECTED SCHOLARSHIPS AND AWARDS

Research Travel Grant	Apr. 2025
Doctoral Scholarship Award	Sep. 2024
Postgraduate Scholarship Award	Sep. 2022
First Prize of the Smart Audiovisual Track, National AI Application Innovation Challenge	Sep. 2024
Champion of the HCI channel, the 2nd Guangzhou Pazhou Algorithm Competition (Top 15/3307)	Sep. 2023
Scholarship from Nanjing University of Information Science and Technology	Nov. 2021
Scholarship from China Mobile Communications Corporation (CMCC)	Nov. 2020
Scholarship from China Mobile Communications Corporation (CMCC)	Nov. 2019

TEACHING EXPERIENCE

HKUST(GZ)	
Big Data Applications for Business and Government (Spring 2025)	Natural Language Processing and Its Applications (Fall 2024)
Red Bird Challenge Camp (Summer 2022, 2023, 2024)	
NUIST & Reading	
Professional Skills for Mathematicians (Fall 2021)	Numerical Analysis (Fall 2021)
Applied Stochastic Process (Fall 2021)	Mathematical Programming (Spring 2021)

SERVICES & SKILLS

PC	2026, COLM, ICML, CVPR, ICLR, AAAI, ACM MM 2025, WWW, AAAI, ACM MM, KDD, NeurIPS, ICML, IJCAI, ICLR, ICASSP, AISTATS, ICWSM 2024, NeurIPS
Review	ACM Computing Surveys; IEEE Transactions on Circuits and Systems for Video Technology; Transactions on Machine Learning Research (TMLR)
Campus	Student Representative, Committee on Undergraduate Studies, HKUST(GZ) Senate (2024–2025)
Stack	Python; PyTorch; Transformers; LLaMA Factory; vLLM & SGLang; LLM MCP&Agents; Web Crawling
Languages	Chinese; English (IELTS 7)